5.18 Ambient Air Quality Monitoring programme (AAQM)



Under the Ambient Air Quality Monitoring (AAQM) programme the Board continued to monitor ambient air quality at Ahmedabad, Vadodara, Surat, Vapi, Bharuch and Rajkot. The ambient air quality samples were collected as per the standard norms for ambient air quality monitoring. The parameters determined during analysis included PM_{10}, $PM_{2.5}$, SO_2, NO_X, NH_3, O3, CO, Lead, Arsenic, Nickel, Benzene and Benzo-A-Pyrene. The data pertaining to the ambient air meteorological data were regularly submitted to the CPCB [Central Pollution Control Board] for compilation and preparation of report. Under the Ambient Air Quality Monitoring (AAQM) programme the Board continued to monitor ambient air quality under other various projects like SAMP, NAMP, and Air Action Plan etc. in the State.

1. National Air Quality Monitoring programme (NAMP) (Annual Average 2022-2023)



Under this project Ambient Air Quality monitoring quarried out at 38 stations in the state with the financial help of the Central Pollution Control Board, Delhi. The ambient air quality samples were collected as per the standard norms for ambient air quality monitoring prescribed by CPCB. The parameters determined during analysis included PM10, PM2.5, SO2, NOX, NH3, O3, CO, Lead, Arsenic, Nickel, Benzene and Benzo-A-Pyrene. Meteorological Data are also monitored at the same locations. Collected data pertaining to the ambient air quality are regularly submitted to the CPCB [Central Pollution Control Board] for compilation and preparation of report.

Sr. No.	City/ Town	Location of Monitoring Station	Parameters			
			PM_{10}	$PM_{2.5}$	SO_2	NO_X
1.	Ahmedabad (City)	Cadila Laboratory Narol	104.0	23.0	15.8	20.9
2.	Ahmedabad (City)	L.D.Engineering Collage	94.0	21.0	13.2	17.6
3.	Ahmedabad (City)	Shardaben Hospital, Saraspur	104.0	23.0	13.5	17.9
4.	Ahmedabad (City)	R.C.Technical School	105.0	23.0	13.2	17.6
5.	Ahmedabad (City)	Behrampura Referal Hospital	102.0	23.0	14.6	19.1
6.	Ahmedabad (City)	Gujarat Labour Welfare Board" Shramyogi Kalyan Bhavan" G-Colony, Sukhramnagar, A'bad	107.0	24.0	16.7	21.2
7.	Ahmedabad (City)	Dyno Wash, 27, Illaben Estate, Narol	141.0	33.0	17.3	22.6
8.	Ahmedabad (City)	Sola L.T. Chankyapuri, Pumping Station, Nr. Kargil Char Rasta	112.0	24.0	14.2	18.9
9.	Bharuch	DahejSez,Dahej	110.0	26.0	17.6	22.5
10.		Sarasvati Township R.I.L.,Dahej	100.0	23.0	13.9	18.4
11.	Ahmedabad (East) Vatva	Naroda police chowkiNaroda GIDC	105.0	23.0	13.5	17.7
12.	Vapi	GEB GIDC, Club 3 Building, Vapi.	107	36	23.2	26.6
13.		Nagar Palika, Vapi.	98	33	20.2	23.5
14.		Yasho Industries, Phase-III, GIDC	108	36	24.2	27.5
15.	Surat	Svr College	99.58	34.08	21.80	25.48
16.		Brc, Udhana (DarshanProcessors)	105.74	36.08	25.96	29.30
17.		Air India Building	101.00	34.42	20.93	24.78
18.		Plot No.413, GIDC, Pandesara	110.00	37.50	25.63	27.62
19.		Plot No. 221, GIDC, Pandesara	115.50	38.25	26.23	29.62
20.		Chaltan Sugar	99.25	33.33	19.60	22.91
21.		M/S CETP of New Palsana Ind.	103.00	35.58	22.69	26.18

22.	Ankleshwar	Rallis India,Ankleshwar	110	36	22.7	26.2
23.		Durga Traders, Ankleshwar	100	34	21.2	24.5
24.		Panoli Industrial Association GIDC Panli	101	34	21.5	24.9
25.		GIDC office, GIDC Jhagadia	108	38	26.1	28.5
26.	Vadodara	Nandesari CETP	111	29	14.50	19.18
27.		GPCB Office	99	25	13.04	17.37
28.		Dandiya Bazar	108	28	14.04	18.58
29.		Sterling Gelatin-Karakhadi	118	30	14.96	19.67
30.		Manjusar Savali	114	29	14.79	19.35
31.	Rajkot	Near Sardhara Corpn., Rajkot.	110	29	14.5	19.5
32.		GPCB Office, Rajkot.	99	26	13.3	17.7
33.	Jamnagar	Fisheries Office, Jamnagar.	103	26	13.8	18.3
34.	Kutch - EAST	Amrut Chamber, Baroi Road, Mundra, Kutch	103	27	13.4	18.0
35.	Morbi.	Eagle Sanitary wares, Village Jambudiya, Morbi.	114	30	14.5	19.5
36.	Morbi.	Morbi Nagarpalika, Morbi.	99	26	13.1	17.5
37.	Ahmedabad (Rural)	M/s Zydus infrastructure Pvt Ltds.(CETP)-Pharmez	104	27	13.6	17.2
38.	Sarigam	Tristar Intermediate (Unit-II), Plot No. 3321, GIDC-Sarigam	94	32	19.4	22.8

*All Parameters are express in **µg/Nm^3**

2. State Air Quality Monitoring programme (SAMP)

Under this project Ambient Air Quality monitoring is carried out at 24 stations in the state. The ambient air quality samples were collected as per the standard norms for ambient air quality monitoring prescribed by CPCB. The parameters determined during analysis included PM_{10}, $PM_{2.5}$, SO_2, NOx, NH_3, O_3, CO, Lead, Arsenic, Nickel, Benzene and Benzo-A-Pyrene. Meteorological Data are also monitored at the same locations.

(Annual Average 2022-2023)

Sr. No.	City/ Town	Location of Monitoring Station	Parameters			
			PM_{10}	**$PM_{2.5}$**	**SO_2**	**NO_X**
1.	Ahmedabad	Jodhpur Shraddha Pumping Station, Satellite Area,	107.0	24.0	14.5	19.0
2.	Ahmedabad	On the Terrace of Behra-Munga	109.0	24.0	13.9	18.3

		School, Aashram Road				
3.	Ahmedabad (East) Vatva	Mukesh Industries, Narol	145.0	34.0	17.7	23.1
4.		Via Hall, Gidc Vatva	128.0	29.0	16.3	21.5
5.		H.P Petrol Pump	105.0	23.0	14.3	18.8
6.	Vapi	GIDC Estate,Vapi	102	34	21.4	24.6
7.		Gpcb Office ,Vapi	99	34	21.1	24.3
8.	Morbi	Atul Pharma, Wankaner GIDC, Wankaner.Morbi.	110	29	14.2	19.0
9.		Sonet Ceramic, Nr. Trajpar Chowkadi, Morbi.	105	27	13.7	18.4
10.	Surat	Delhi Gate Police Chowki	102.58	35.42	22.38	25.89
11.		Hi-Choice Processers, Sachin	107.08	37.00	23.08	26.48
12.		Garden SilkMills, Kadodara	103.25	36.67	21.45	24.80
13.	Ankleshwar	PiramalVill.Ankleshwar	98	33	19.5	22.9
14.		ValiaRoad,Ankleshwar	98	33	19.6	22.9
15.	Vadodara	Bapod,Vadodara	102	26	16.31	17.59
16.		Chhani,Vadodara	100	25	13.09	17.42
17.		Gotri,Vadodara	104	26	13.12	17.78
18.	Rajkot	Mahangarpalika pumping Station , Rajkot	104	27	13.7	18.4
19.	Jamnagar	GPCB Office,Jamnagar	96	24	13	17.4
20.	Bhavnagar	Alang-ship Breaking Yard, Alang	113	25	16.2	21.6
21.		ChitraGIDC,Bhavnagar	123	29	17.1	22
22.		Sosiya Ship Breaking Yard,Bhavnagar	113	26	15.9	21.2
23.	Kutch - EAST	Lunwa-Bhachau,Gandhidham belt	111.0	29.0	15.3	20.4
24.	Kutch - EAST	Gandhidham-Kandlabelt	98.0	25.0	13.0	17.5

*All Parameters are express in **µg/Nm^3**

3. Volatile organic compound (VOC) Monitoring

Gujarat Pollution Control Board had started monitoring of VOC at 25 Clusters in Gujarat. VOC monitoring is carried out once in a month at each location for 24 hrs. as per CPCB guidelines

Yearly Average 2022-2023																
Sr. No	**Monitoring Location/ Voc Clusters**	**Methanol**	**Ethanol**	**Acetone**	**Iso-propyl Alcohol**	**Methylene di-chloride**	**Butanol**	**Choloro- form**	**Carbon tetra chloride**	**Benzene**	**Methyl Isobutylketone**	**Toulene**	**p-xylene**	**m-xylene**	**o-xylene**	**Ethyl Methyl Ketone**	**Benzo (a) pyrene**
1.	Swami Shanti Prakash Pvt.Ltd. Plot No.9, 10, Mahagujarat Industrial Estate, Moraiya, Ta. Sanand, Dist.Ahmedabad	9.50	1.99	1.78	2.20	1.45	2.10	1.24	1.19	3.53	2.36	3.77	2.56	2.59	2.65	2.42	<1.0
2.	Loxim Industries Ltd. Vill:Iyava, Ta. Sanand Dist.Ahmedabad	9.37	1.79	2.04	2.12	1.39	2.13	1.25	1.16	3.38	2.39	3.63	2.64	2.62	2.56	2.33	<1.0
3.	Vatva Industrial Association, GIDC - Vatva	9.48	1.91	1.99	2.2	1.36	2.19	1.21	1.16	3.44	2.36	3.69	2.65	2.65	2.64	2.39	<0.1
4.	Udhyog Nagar Police Chwki, GIDC- Naroda	9.58	1.95	2.06	2.25	1.48	2.07	1.26	1.19	3.41	2.46	3.67	2.75	2.64	2.59	2.45	<0.1
5.	Odhav Industrial Asociation, L-203, GIDC – Odhav	9.49	2.01	2.05	2.13	1.64	1.95	1.2	1.13	3.37	2.4	3.6	2.57	2.58	2.57	2.45	<0.1
6.	Gangotri,Plot No:03,Anjar Kutch	8.93	1.35	1.56	1.76	1.34	1.85	1.17	1.12	2.99	2.03	3.14	2.15	2.19	2.19	2.0	<0.1
7.	Bhachau-Gandhidham Road,Opp.Gujarat NRE Coke Div., Lunva Kutch	8.79	1.41	1.55	1.82	1.30	1.67	1.20	1.11	2.97	1.74	3.03	2.04	2.10	2.10	2.0	<0.1
8.	Amrut Chambers, Baroi Road, Mundra. Kutch	8.79	1.42	1.53	1.74	1.32	1.61	0.82	0.90	2.86	1.79	2.89	2.13	2.12	2.02	2.02	<0.1
9.	Rallis India Ltd. Ankleshwar	10.27	2.19	2.63	2.38	2.19	2.49	1.69	1.66	3.57	2.40	3.82	3.11	3.03	3.09	2.71	<0.1
10.	Hikal Ltd, GIDC – Panoli	9.84	1.95	2.29	2.31	2.02	2.49	1.42	1.34	3.51	2.48	3.82	3.04	2.92	2.85	2.59	<0.1
11.	DCM Shreeram Limited, Jagadia Industrial Estate.	9.78	2.01	2.17	2.19	1.46	2.41	1.34	1.24	3.49	2.47	3.72	2.82	2.86	2.73	2.37	<0.1

12.	Gujarat Alkalis & Chemicals Ltd, Dahej Area, Bharuch	9.35	1.59	1.80	2.05	1.44	1.93	1.30	1.18	3.35	2.26	3.50	2.69	2.69	2.59	2.29	**<0.1**
13.	Plot No: 279, GIDC-Chitra, Bhavnagar	8.90	1.43	1.48	1.62	1.22	1.57	1.13	1.14	3.01	1.93	3.23	2.15	2.20	2.12	1.99	<0.1
14.	Sairam Industries Ltd, Chhatral GIDC, Dist.: Gandhinagar	8.81	1.49	1.55	1.77	1.29	1.74	1.18	1.11	3.01	1.90	3.17	1.96	2.01	1.83	1.76	<0.1
15.	Hathubha Chudasama's Terrace Near Water Tank, Moti Khavdi (VOC)	8.85	1.58	1.55	1.89	1.29	1.67	1.19	1.11	3.04	1.82	3.23	2.10	2.13	1.91	1.81	<0.1
16.	Amul Industries Ltd, Nr. Sardhara Corp. Aji Industrial area, GIDC, Rajkot	9.03	1.54	1.62	1.79	1.30	1.65	1.14	1.13	2.95	1.90	3.21	2.11	2.22	2.16	1.96	<0.1
17.	Mecloid Pharma GIDC sarigam	8.86	1.31	1.28	1.63	1.25	1.38	1.16	1.10	2.81	1.66	2.78	1.87	2.20	2.07	2.18	<0.1
18.	High Choice Proceser Pvt. Ltd.,Plot No. 264, Road No.2,GIDC – Sachin, Surat	9.62	1.73	1.94	2.15	1.57	2.29	1.31	1.25	3.31	2.43	3.56	2.79	2.77	2.55	2.44	<0.1
19.	Ichchhapor Village , Hajira, Surat	9.71	1.69	1.76	2.12	1.96	2.37	1.49	1.38	3.36	2.62	3.64	3.04	2.80	2.81	2.52	<0.1
20.	Pandesara GIDC – Office,Surat	9.78	1.83	2.35	2.23	1.95	2.36	1.58	1.42	3.43	2.46	3.75	3.11	3.02	2.76	2.62	<0.1
21.	ECPL PumpHouse, Chikuwadi, ECP Office, Nr. ICPL, Vadodara	9.35	1.73	1.87	2.04	1.46	2.19	1.21	1.15	3.27	2.24	3.47	2.43	2.52	2.38	2.18	<0.1
22.	Lubrizol Advance Material Pvt. Ltd. Savli, Vadodara	9.54	1.87	1.86	2.05	1.44	2.20	1.25	1.17	3.20	2.33	3.41	2.71	2.67	2.67	2.25	<0.1
23.	Above Mahuvad Fire Station, Nr. Transpack Ind. Ekalbara, Vadodara	9.40	1.90	1.84	2.24	1.51	2.02	1.20	1.21	3.48	2.29	3.71	2.73	2.66	2.61	2.35	<0.1
24.	Nandesari CETP, Plot NOI. 153, GIDC – Nandesari, Vadodara	9.36	1.87	1.91	2.11	1.27	1.95	1.14	1.09	3.35	2.22	3.59	2.42	2.30	2.27	2.27	<0.1
25.	Club -3 Bldg.Rishita Fashions, Plot No. - 1102, Third Phase,GIDC - Vapi.	8.87	1.46	1.40	1.87	1.34	1.77	1.17	1.15	2.90	1.80	3.16	2.18	2.17	2.02	1.97	<0.1

***Except Benzo (a) pyrene all parameter are expressed in µg/Nm^3 and Benzo (a) pyrene is expressed in ng/m^3**

5.19. COASTAL MONITORING:

Gujarat has 1600 kms long coastal line. Industrial wastewater from the nearby areas is discharged into Coastal Zone through the river and due to residential development in coastal zone domestic wastewater generated from the major cities and Nagarpalikas goes to costal area.

To know the quality of coastal waster, GPCB has started monitoring of coastal water at certain locations on monthly basis

Sr. No.	Sampling Location/Stations	Parameters*				
		pH	DO	BOD	COD	NH3-N
1.	Sea Water at Dhuvaran, Anand	8.30	6.73	1.43	NA	0.82
2.	Estuarine water at Dhuvaran, Anand	8.30	6.46	1.33	NA	0.72
3.	Coastal Water near NearStambheshwarMahadev Temple, Kavi-Kamboi	7.82	6.79	1.43	NA	BDL
4.	Coastal Water near Near Reliance Jetty, Dahej	7.85	6.78	1.34	NA	BDL
5.	Coastal Water near Near Birla Copper (Hindalco) Jetty, Dahej	7.87	6.81	1.28	NA	BDL
6.	Coastal Water near Near LNG terminal jetty Dahej	7.89	6.85	1.29	NA	BDL
7.	River Narmada Estury at Golden Bridge Bharuch	8.08	7.02	0.92	9.25	BDL
8.	River Narmada Estury at Bhadbhut	8.06	6.75	1.22	12.5	BDL
9.	Sea water sample at village Mandvi.Bandar road, Mandvi.	7.90	5.31	2.73	76	0.54
10.	Coastal water (Surface sea water sample at Kandla)	7.95	5.09	2.59	NA	0.54
11.	Water sample at Vill- Kidana under coastal monitoring project.	7.81	0.28	33.33	NA	35.24
12.	Coastal water sample of Kandla port Opp. Marine Bhavan ,at Kanda	7.95	5.09	2.60	NA	0.54
13.	Domestic water sample of outlet of Ksez/Gandhidham city Nr Kidana	7.81	0.30	33.33	NA	30.97
14.	Sea water sample at New jetty of Mundra port	7.95	5.10	2.35	NA	0.58
15.	Sea water sample at Old jetty of Mundra port	7.96	5.13	2.76	NA	0.56
16.	Sea water of confluence point of Adani & Tata thermal power plant	7.94	4.90	2.87	NA	0.56
17.	Sea water sample at Bhadreshwar, Tal Mundra, Kutch	7.99	5.02	2.60	NA	0.56
18.	Excel Crop Care Ltd., Bhavnagar	7.49	4.44	35.7	163	9.8
19.	BMC waste water at Kansara nalla, Bhavnagar	7.57	BDL	96.14	336	11.53
20.	Pipavav Port, Amreli	8.1	6.35	3.41	43.6	2.99
21.	Nirma Ltd, Kalatalav, Bhavnagar	8.12	6.35	3.27	38.5	3.78

22.	Plot 6, SBY,Alang, Bhavnagar	8.11	6.3	4.68	48.9	4.013
23.	Plot 110, SBY, Sosiya, Alang, Bhavnagar	8.11	6.36	4.47	46.5	3.55
24.	JMC Domestic out fall w/w from pumping station of Gandhinagar Rly station	7.36	0	69	194	7.37
25.	Sea water sample at Rozy Port, Near Light House	7.90	6.0	1.1	NA	0.56
26.	Sea water sample at Sachana Ship Breaking yard	7.98	5.1	1.3	NA	0.56
27.	Final outlet of Tata chemicals mithapur (after dilution after settling)	7.97	5.2	1.1	NA	0.56
28.	Sea Water Sample Nr. Vill. Arambhada down steam of Tata Chemicals Ltd**.**	7.90	5.2	1.1	NA	0.56
29.	Relience Industries Limited (Sea water Basin Outlet), Jamnager	7.91	5.2	1.1	NA	0.56
30.	Sea water sample collected @ 2 KM away from RIL jetty, Daman	7.93	5.1	1.1	NA	0.56
31.	Essar Oil Limited(Sea water Basin Outlet), Gandhinager	7.97	4.3	1.1	NA	0.56
32.	Sea water sample collected @2 KM away from Essar Jetty, Daman	7.95	5.1	1.1	NA	0.56
33.	Cooling water return through open channel water sample of GEB TPS, Sikka	7.93	5.2	1.1	NA	0.56
34.	Sea water sample at Narara Island	7.91	5.2	1.2	NA	0.56
35.	Surface Sea Water Sample of Muldwarka, Kodinar, Dist: Gir Somnath	7.99	5.5	2.15	40.73	0.28
36.	Surface Sea Water Sample Nr. Somnath Temple, At: Veraval, Dist: Gir Somnath	8.01	5.56	2.15	41.5	0.30
37.	Surface Sea Water Sample at Holyday Camp, Choravad	7.99	5.53	2.14	39.56	0.28
38.	Surface Sea Water Sample Nr. Indian Rayon & Ind. Ltd. Veraval	8.05	5.55	2.18	43.60	0.30
39.	Surface Sea Water Sample Nr. Intekpoint of GHCL, Sutarapada	8.00	5.54	2.16	41.14	0.30
40.	Domestic W/W of. Veraval Municipality Nr. Vasant vihar Engg., Veraval	8.11	BDL	207	640	28.15
41.	W/W Sample of Fish Processing Unit, GIDC, Veraval	8.03	0.1	289.6	907	96.04
42.	Surface sea water sample at Navalakhi Port, Navalakhi, Tal. : Maliya, Dist. : Morbi.	8.04	5.54	2.16	46.41	0.44
43.	Surface sea water sample at Surajbari Creek, Surajbari, Tal. : Maliya, Dist. : Morbi.	8.03	5.48	2.21	45.00	0.44
44.	Sample from sea at chopati porbandar	7.58	3.96	0.95	NA	0.56

45.	Domestic water sample of nagarpalika at jubeeli bridge, porbandar	7.39	NA	67	NA	0.93
46.	Sarigam Diffuser at – Tadagam, Di-Valsad.	7.90	5.94	1.3	NA	0.96
47.	River Tapi at Umara Temple	7.72	6.11	1.95	NA	0.79
48.	River Tapi at KRIBHCO Jetty	8.02	6.45	1.74	NA	0.83
49.	River Tapi at ESSAR Jetty	8.03	6.34	1.79	NA	0.85
50.	E.C.P. at 'J' Point near Vill. Sarod, Bharuch	7.87	NA	65.3	353	11.95
51.	Coastal Water Sample at Vill. Sarod, Bharuch	8.11	6.73	1.39	NA	1.67
52.	Estuarine Water Sample at Vill. Sarod, Bharuch	8.14	6.67	1.36	NA	1.68
53.	Estuarine water of river Damanganga at Daman Jetty	7.68	6.15	1.175	NA	0.7225
54.	Estuarine water of river Damanganga at Kanchigam, Daman	7.66	5.42	1.58	NA	2.89
55.	Estuarine water of river Kolak at Kolak Jetty, Daman	7.89	6.19	1.02	NA	0.60
56.	Estuarine water of river Par at Umarsadi Jetty, Daman	7.89	6.39	1.03	NA	0.75

***All parameters are express in mg/l except pH. (NA- Not Analysed (Due to High Chloride))**